Neonatal Intensive Care Unit Text Summarization

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Group 10

BACKGROUND

Motivation & Clinical Need

Current Challenges

NICU nurses face significant administrative burdens due to manual charting and patient handovers. Documenting and patient data consumes considerable time, reducing nurses' availability for direct patient care

Parental Stress

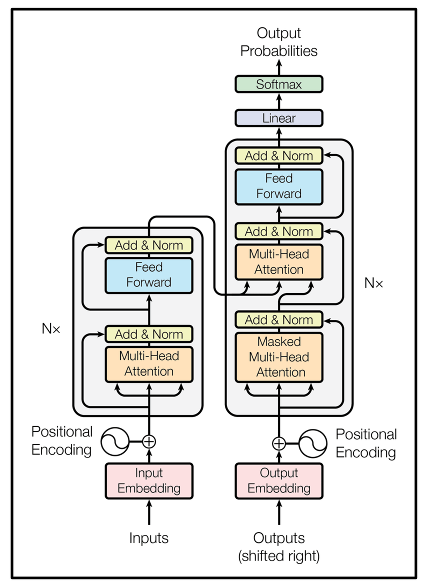
Parents of NICU infants frequently experience anxiety and emotional strain from limited and unclear communication about their child's health. Traditional clinical updates often lack clarity and timeliness, leaving parents uncertain and anxious, highlighting a critical need for empathetic, real-time communication

Impact

Manual summarization and ineffective handovers significantly heighten the risk of miscommunication, errors, and oversights, potentially leading to adverse neonatal outcomes. Accurate, timely communication is vital for patient safety, making the improvement of information transfer an essential aspect of NICU patient care quality.

Chatbot

Language Model



Transformer Architecture

Neonatal Vital Sign Fluctuations

Common Vital Sign Fluctuations

NICU patients frequently exhibit fluctuations in vital signs, including episodes of bradycardia, tachycardia, tachypnea, and bradypnea These fluctuations can be transient or indicate underlying medical issues requiring intervention

Early Warning Signs

Variations in vital signs often serve as crucial early indicators of neonatal health conditions

Clinical Importance

Rapid detection and accurate reporting of vital sign changes are essential for timely interventions, ensuring patient safety and improving outcomes

Generative AI & Large Language Models (LLMs)

Generative AI has significantly advanced natural language processing (NLP), enabling automated summarization and streamlined medical documentation.

- **Healthcare Applications:** Language models support clinical tasks including automated patient summaries, EHR integration, and empathetic chatbot interactions for patient communication.

Figure 1: Illustration depicting a NICU nurse monitoring and caring for an infant, highlighting the critical importance of continuous neonatal care and timely intervention.
Image generated with Gemini 2.0 Flash

OBJECTIVES

- Develop a neonate patient vital sign simulator for realistic neonatal health data and validation.
- Using LLMs to
  - Automate NICU charting documentation to reduce clinician workload.
  - Enhance patient handovers with AI-driven text summarization of patient data
  - Develop a nurse chatbot for specific patient queries
  - Provide real-time parental updates via a chatbot.

METHODOLOGY – NICU Vital Sign Patient Simulator

Datasets

- *Preterm Infant Cardio-Respiratory Signals Database*
- *Model for Simulating ECG and PPG Signals with Arrhythmia Episodes*

The following datasets were sourced from Physionet.org

Dataset Name	Content	Number of Records	Parameters	Sampling Frequency	Duration
PICSDB	ECG and respiration signals from preterm infants	10 infants	Post-conceptional age, birth weight, ECG & respiration sampling frequency	ECG: 250-500 Hz, Respiration: 50 Hz	Up to 72 hours per infant
ECG-PPG Sim	Simulated ECG and PPG signals with arrhythmias	10 Simulations	Arrhythmia type, signal type, duration, sampling frequency	ECG: 300 Hz, PPG: 60 Hz	600 seconds per simulation

Table 2: NICU Patient Simulator Datasets: The PhysioNet PICSDB dataset provides up to 72 hours of real ECG (250–500 Hz) and respiration (50 Hz) recordings from ten preterm infants including post-conceptional age and birth weight. In contrast, the ECG-PPG Arrhythmia Simulator dataset generates 600-second synthetic ECG (300 Hz) and PPG (60 Hz) signals that model arrhythmias such as bradycardia, tachycardia, and atrial fibrillation for controlled testing.

Data Processing & Preprocessing

- Extract Data: Load ECG & Respiration (WFDB)
- Identify Peaks: R-peaks (HR) & Respiration peaks (RR)
- Segment Data: 8-hour subsets (NICU shifts)
- Filter Outliers: IQR-based cleaning
- Down sample: 5-min intervals (96 points)

Synthetic Data Generation

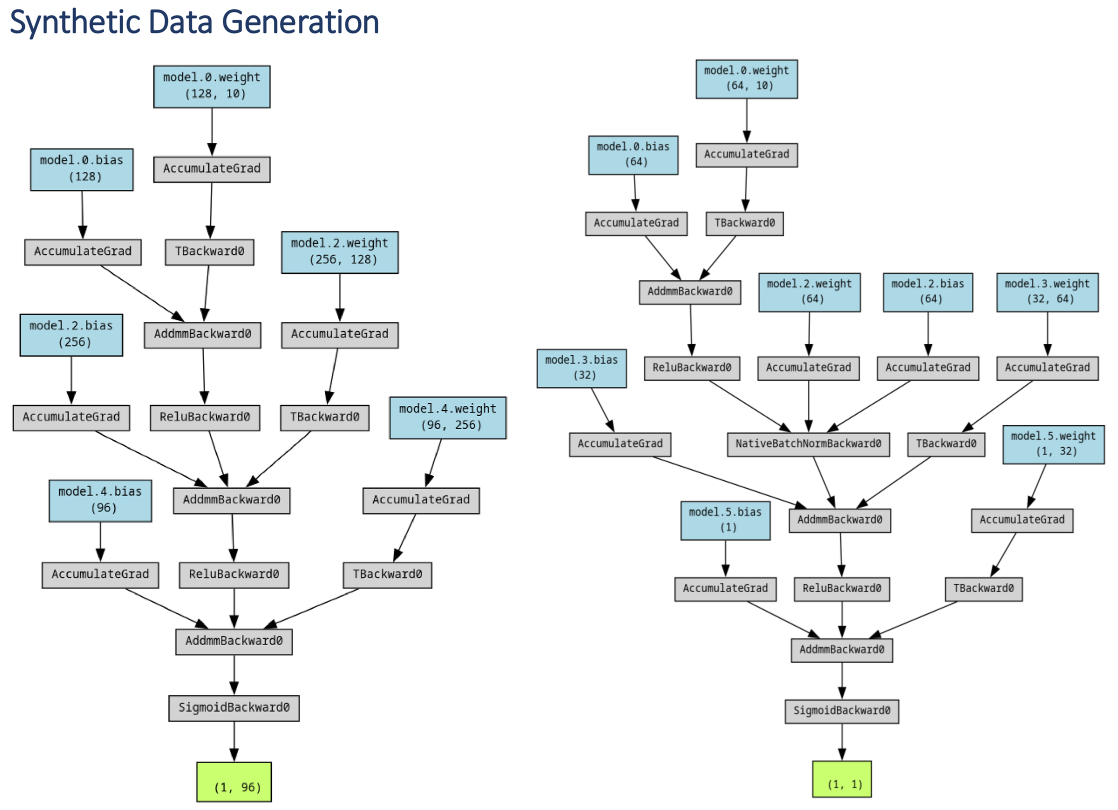


Figure 3: Computational Graph of the GAN Generator. The architecture comprises fully connected layers with ReLU activations and a final sigmoid output that maps random noise to either a 96-point synthetic sequence for heart and respiratory rates or a single-dimensional synthetic heart rate reading. Trained on real neonatal data, the generator produces realistic vital sign patterns.

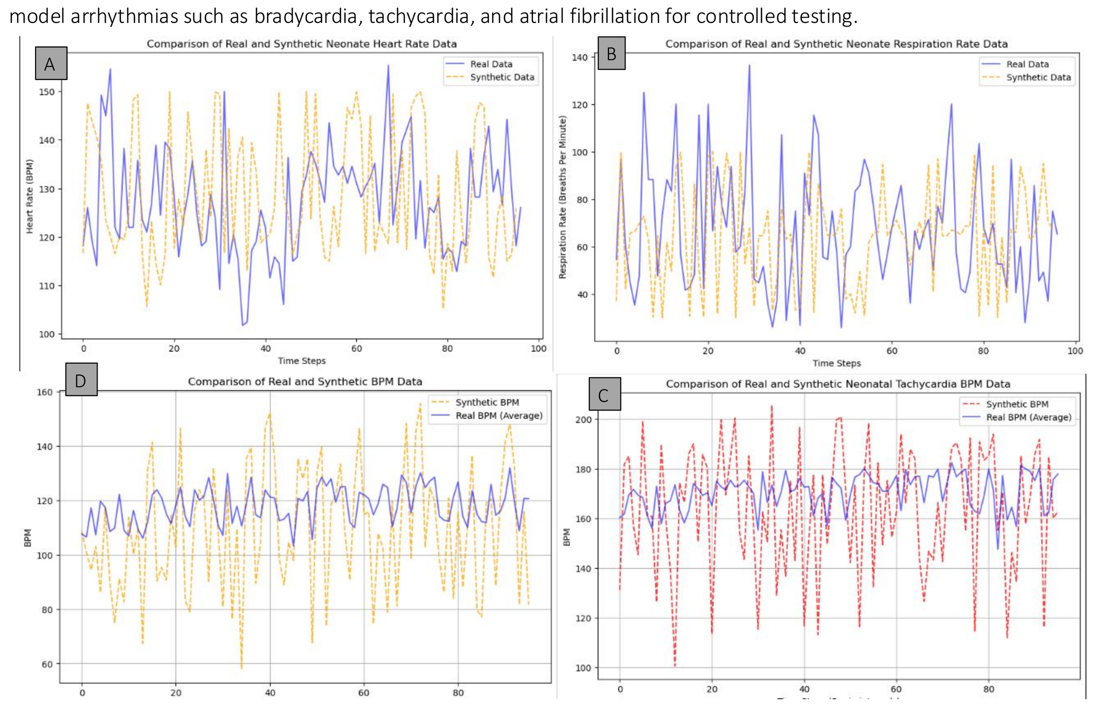


Figure 4: GAN Comparison of Neonatal Data: Panel A shows real versus synthetic heart rate (BPM) fluctuations over time. Panel B compares real and synthetic respiration rate (breaths/min). Panel C overlays average real and scaled synthetic bradycardia data, highlighting the downward adjustment during bradycardic episodes. Panel D illustrates upscaled synthetic tachycardia data that capture extreme heart rate fluctuations, demonstrating the GAN model's ability to simulate physiologically relevant pathological conditions.

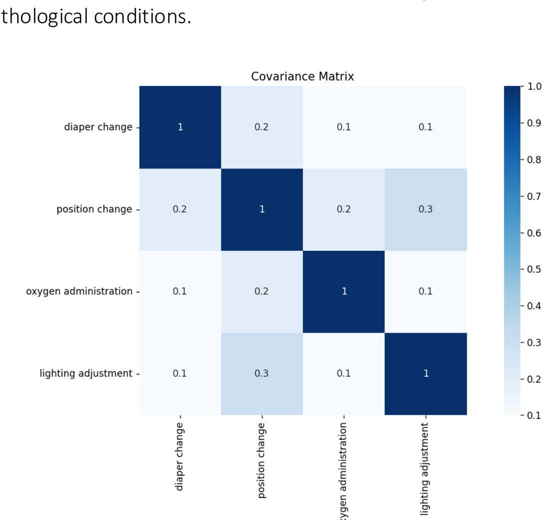
Data Scaling

Condition	Scaling Method
Bradycardia	Downscale HR <100 BPM, neonatal range 50-85 BPM
Tachycardia	Upscale HR >160 BPM, neonatal range 160-220 BPM

Table 1: GAN Scaling Adjustments: Heart rates below 100 BPM are downscaled to 50–85 BPM (bradycardia), while rates above 160 BPM are upscaled to 160–220 BPM (tachycardia).

Intervention Modeling

- Scheduled: Fixed NICU routines (feeding, repositioning)
- Unscheduled: Random events (Poisson distribution)
- Co-occurring Interventions: Covariance-based relationships
- Time-Based Adjustments: Adjustments for day/night cycles
- Pain Management: HR, RR, Temp deviations → Weighted scoring



METHODOLOGY – LLM-based Text Summarization Systems

Database Implementation

Utilized SQLite & SQLAlchemy for storage and retrieval of user-chatbot interactions and patient summaries, to ensure session continuity.

Evaluating LLM Generated Responses

Automated Evaluation: Generated responses are assessed using standardized criteria for *Relevance* and *Groundedness*.

Manual Evaluation: Scoring outputs with custom metrics, such as *Answer Specificity Score*, *Hallucination Count* & *Recall*.

We also implement custom Likert Scales in tandem with A/B testing to score the tone and contextual appropriateness of the responses.

Evaluator	Metric Explanation
Groundedness	Gauges alignment with reference context
Relevance	Indicates how well the answer addresses the query
Answer Specificity Score	Measures alignment with the requested subset
Hallucination Count	Number invented facts
Recall	Reflects how many relevant details were retrieved

Table 3: LLM Evaluator Explanation

Criterion (A)	1 (Poor)	2 (Fair)	3 (Good)	4 (Very Good)	5 (Excellent)
Comprehensiveness: Does the summary cover all critical patient data?					
Accuracy: Are all details precise and free of errors?					
Clarity: Is the summary structured clearly and easy to follow?					
Relevance: Are all details actionable and clinically important?					

Criterion (B)	1 (Poor)	2 (Fair)	3 (Good)	4 (Very Good)	5 (Excellent)
Clarity: Is the update easy to understand, free of jargon?					
Relevance: Does the update include only important and relevant details?					
Tone: Is the tone empathetic and reassuring?					
Conciseness: Is the update brief yet informative?					

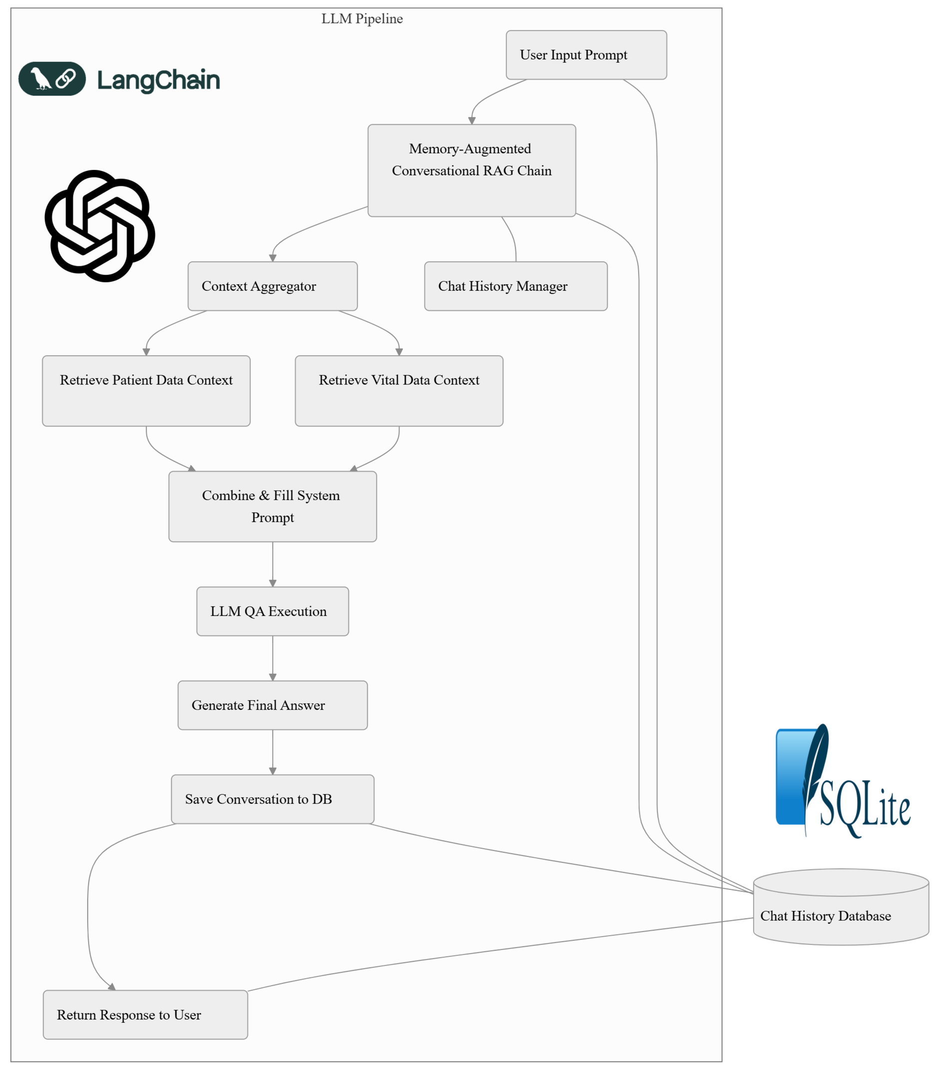
Table 4: Likert Scale for evaluating Nurse Summarizer & Chatbot (A). Likert Scale for evaluating Parent Chatbot (B)

LangSmith for LLM Evaluation

LangSmith is a development framework that can be used to trace LLM applications and leverage LLMs to automatically score generated responses

Text Summarization/Chatbot Pipeline

The LLM systems (chatbots, summarizer & auto charter) are built from the same underlying pipeline.



System Prompting

System prompts provide clear instructions and context to guide/tailor LLMs in generating responses given a particular context. In our case: the system prompts instruct the LLM on how to interpret patient data, assess vital sign conditions, and communicate effectively. This ensures that outputs are clinically accurate, contextually appropriate, and consistent with intended clinical communication standards.

"You are a nurse in the NICU at the end of your shift, preparing for patient handover."

"Provide the incoming nurse with all pertinent information for their shift in a concise, technical summary."

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Results

Generative Adversarial Network Evaluations

Normal Heart Rate Validation

(A) Metric	Real Data	Synthetic Data
KS Statistic	0.15	0.12
KS Test p-value	0.39	0.47
Mean /STD	135.4/12.3	137.8/14.1

Normal Respiration Rate Validation

(B) Metric	Real Data	Synthetic Data
KS Statistic	0.13	0.10
KS Test p-value	0.41	0.52
Mean /STD	45.2/8.1	44.9/9.3

Table 5: Statistical Evaluation of GAN-Generated Neonatal Vital Signs. The tables present validation metrics comparing real and GAN generated synthetic data for neonatal heart rate (A) and respiration rate (B).

Tachycardia Validation

(A) Metric	Real Data	Synthetic Data
# of Episodes	57	62
KS Statistic	0.24	0.21
KS Test p-value	0.034	0.068

Bradycardia Validation

(B) Metric	Real Data	Synthetic Data
# of Episodes	42	45
KS Statistic	0.11	0.09
KS Test p-value	0..49	0.53

Table 6: Validation of GAN Generated Tachycardia (A) and Bradycardia (B) Episodes.

LLM Application Evaluation

(A) Evaluator	Score
Groundedness	0.907
Relevance	0.991
Answer Specificity Score	0.749
Hallucination Count	0.15
Recall	0.910

(B) LLM Tool	Average Likert Scale Score
Nurse Chatbot	3.75
Nurse Summarizer	4.25
Parent Chatbot	4.5
Auto Charter	4.5

Table 7: Evaluation metrics for the Nurse Chatbot, averaged over 6 experiments (10 runs each (A). Average Likert-scale scores (out of 5) for each LLM-based NICU tool, reflecting combined ratings of comprehensiveness, accuracy, clarity, relevance, tone, and conciseness (B).

Example Generated Patient Summary

Patient: Jonathan Donathan, male, preterm (approximately 31 weeks post-conceptual age, 2.5 kg). No underlying conditions.

Overview of the shift:
- Vital signs were obtained at multiple time points with notable variability.
- Heart rate readings ranged from 119 to 149 bpm. Several measurements (e.g., 119–136 bpm) were below the expected lower threshold for preterm infants, indicating episodes of bradycardia, while a few values (141–149 bpm) were within range.
- Respiratory rate values varied from 33 to 87 breaths/min. Some readings (33–35 breaths/min) fell below the expected range (bradypnea), whereas others (76–87 breaths/min) exceeded it (tachypnea); thus, both abnormal conditions were observed.
- Body temperature ranged from approximately 36.6°C to 37.4°C. Although many readings were near the acceptable range for a preterm infant, a number exceeded the upper limit.
- Throughout the shift, the patient's pain level remained low.

Interventions performed included scheduled diaper changes and several position changes (notably left lateral, right lateral, and prone) as well as a family visitation.

This summary should help in understanding the variability in vital signs and the interventions carried out during the shift.

↑

Normal preterm baby occasionally experiencing certain abnormal conditions

Conclusion

- GAN-based models effectively replicate vital sign patterns
  - HR, RR, Bradycardia & Tachycardia
- LLM-driven summarization and chatbot tools effectively streamline communication
- Continued testing, evaluation, refinement required before deployment
- Reduces administrative burdens, supports timely interventions, and enhance patient care in the NICU

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